

CreditCardPayment.java - Explanation

ALGORITHM / EXECUTION FLOW:

1. Constructor takes in both the base data (amount) and card data.
2. It passes the base data UP to the parent class via super().
3. It overrides the process() method to validate the card specifically.

SECTION 1: SPECIFIC STATE (Encapsulation)

These fields only exist for Credit Cards. PayPal doesn't have a CVV.

SECTION 2: THE CONSTRUCTOR

Calls the parent (Payment) constructor to set up ID, time, and amount, then saves the credit card specific data.

SECTION 3: THE OVERRIDDEN ALGORITHM

This is Polymorphism in action. When PaymentProcessor calls process(), if the object is a CreditCard, THIS block of code runs. It validates the CVV algorithmically, simulates network authorization, and handles success or failure states independently.